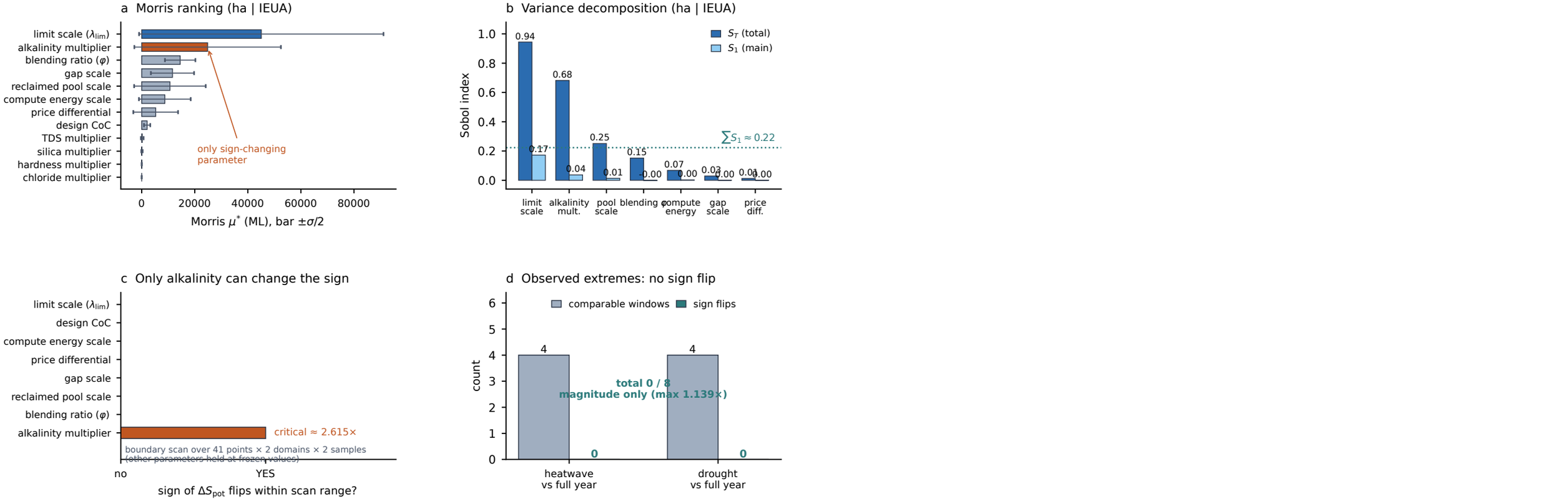

Data: A34_WP5/wp5_morris.csv, wp5_sobol.csv, wp5_extreme_cases.csv, wp5_extreme_summary.json.

Morris/Sobol: Saltelli design on config ha|IEUA; LADWP configs report NOT_IDENTIFIABLE (6-7 of 512 points fail the gate; NaN silently propagated) and are NOT plotted. Extremes: n_eff = 64 of 64 windowed scenarios after the dual gate. ZERO CONVENTION: |dS_pot| < 1e-6 ML counted as 0.

LIMITATION: LADWP Sobol indices are not identifiable and are omitted rather than imputed. The alkalinity critical value ~2.615x has NO observational support (measured samples: IEUA 140 mg/L, LADWP 7.1 mg/L), so it is a sensitivity boundary, not a predicted regime. Extreme-event windows use driver-only forcing and a single-city domain.